J2
TP CLOCKS
CLK_INIT_P 1 2 CLK_INIT_N
CLK_SAMP_P 3 4 CLK_SAMP_N
COMP_OUT_P 5 6 COMP_OUT_N
CLK_LOGIC_N 7 8 CLK_LOGIC_P
CLK_COMP_N 9 10 CLK_COMP_P
11 12
GND GND

DisplayPort Molex 472720024
J1
L3_N 1
GND 2
L3_P 3
L2_N 4
GND 5
L2_P 6
L1_N 7
GND 8
L1_P 9
L0_N 10
GND 11
L0_P 12
AUX_P 15
GND 16
AUX_N 17
MH1 MH2 MH3 MH4
VDD
R1 10k
100n C1
GND
GND Earth

Place pull downs further from DUT, and diff resistors closer
R10 100 CLK_COMP_P CLK_COMP_N
R12 100 CLK_LOGIC_P CLK_LOGIC_N
COMP_OUT_N COMP_OUT_P
R11 100 CLK_SAMP_N CLK_SAMP_P
R13 100 CLK_INIT_N CLK_INIT_P
R2 1k R3 1k R4 1k R5 1k R6 1k R7 1k R8 1k R9 1k
GND
TH1
Thermistor_NTC

VOCM_EXT
C2 1u C3 100n
GND
R15 DNP
R16 499
J4
SMA Molex 733910070
1 VIN_P_EXT
2
GND
R17 499
R18 DNP
J3
SMA Molex 733910070
1 VIN_N_EXT
2
GND
Diff: Single-Ended:
J5
DIFF/SE CFG
R14 50
GND
J6
TERM CFG
GND
R19 54.9
R20 54.9
GND GND
50Ω: Open:
GND GND

Buffer: Attenuator:
J7
ATTEN CFG
C4 51pF
R21 499
Enabled: Disabled:
J9
TERM CFG
VDD_DAC
R23 1k
R24 1k
GND GND
J8
SMA Molex 733910070
1 VOCM_EXT
2
GND
U1
DiffAmp TI THS4541
VOCM
IN+
IN−
PD_N
VS_P
VS+
OUT−
OUT+
NC
NC
VS_N
AMP_PD_N
R22 499
C5 51pF
R25 499
R26 499
C6 100nF

Disable R−divider when measuring VDD_DAC current
J11
TP AMP
VIN_P_EXT 1 2 VIN_P
VIN_N_EXT 3 4 VIN_N
VOCM_EXT 5 6 AMP_PD_N
AMP_PD_EXT 7 8
GND
DiffAmp: Bypass:
J10
VIN CFG
VIN_N_EXT
VIN_P_EXT
JP1 CFG CLAMP
JP2 CFG CLAMP
VIN_N
VIN_P
VDD_DAC
D1 BAV99
D2 BAV99
GND
Disconnect or DNP when measuring VDD_DAC current

Place as empty pads, on the board backside
R27 DNP VDD VDD_A
R28 DNP VDD VDD_IO
R29 DNP VDD VDD_D
R30 DNP VDD VDD_DAC
J12
MiniFitJr Molex 469990014
FB1 FB2 FB3 VDD
J13
POWER
VDD_IO VDD_A VDD_D VDD_DAC VS_P VS_N
GND

VDD_BDAQ should be 2.5V
VDD_BDAQ VDD_BDAQ
2.5V AMP OFF AMP ON 0V
R31 10k
AMP_PD_EXT
Q_PMOS_GSD
Q1
AMP_PD_N
R32 10k Vs−
AMP OFF AMP ON 2.5V
VS_N
VS_N should be −1V

Enabled when PD_N > (Vs−) + 1.7 V
Disabled when PD_N < (Vs−) + 0.7 V
Since thresholds are referenced to Vs− use it for the passive pulldown

U3
ASIC FRIDA 65A
FRIDA65A
VDD_DAC VIN_P VIN_N
VSS_A VDD_DAC VSS_DAC VIN_P VIN_N VSS_D VSS_D
VDD_A VSS_A CLK_INIT_P CLK_INIT_N CLK_SAMP_P CLK_SAMP_N RST_B
VDD_D VSS_D CLK_COMP_P CLK_COMP_N CLK_LOGIC_P CLK_LOGIC_N VDD_IO
SPI_SCLK SPI_SDI SPI_SDO SPI_CS_B COMP_OUT_P COMP_OUT_N VSS_IO PAD
VDD_IO
R36 10k
RST_B
GND GND GND

C7 47u C10 1u C13 100n C16 100n VDD_A GND
C19 47u C22 1u C25 100n C28 100n VDD_IO GND
C8 47u C11 1u C14 100n C17 100n VDD_D GND
C20 47u C23 1u C26 100n C29 100n VDD_DAC GND
C9 47u C12 1u C15 100n C18 100n VS_P GND
C21 47u C24 1u C27 100n C30 100n VS_N GND

VDD_BDAQ
C31 1u C32 100n
GND
J14
RJ45 Molex 446200002
VDD_BDAQ
AMP_PD_EXT
RST_B_EXT
SPI_SCLK_EXT
SPI_SDI_EXT
SPI_CS_B_EXT
SPI_SDO_EXT
GND

U2
TI LevelShifter SN74AXC8T245RJWR
VDD_BDAQ VCCA
RST_B_EXT A1
SPI_SCLK_EXT A2
SPI_SDI_EXT A3
SPI_CS_B_EXT A4
SPI_SDO_EXT A5
A6 A7 A8
RX
TX
DIR1
DIR2
VCCB VCCB
RST_B B1
SPI_SCLK B2
SPI_SDI B3
SPI_CS_B B4
SPI_SDO B5
B6 B7 B8
OE_N
GND GND
VDD_BDAQ
R33 10k R34 10k
R35 10k
DIR1 pull−up −> A1−A4 input
DIR2 pull−up −> A5−A8 output
GND GND

VDD_IO
C33 100n C34 1u
GND

J15
SPI TP
RST_B_EXT RST_B
SPI_SCLK_EXT SPI_SCLK
SPI_SDI_EXT SPI_SDI
SPI_SDO_EXT SPI_SDO
SPI_CS_B_EXT SPI_CS_B
VDD_BDAQ
GND

H1 H3 H2 H4